

PROJECTS

AI Blog Generator | Python, Flask, LLM APIs, Prompt Engineering

Feb 2025 – Mar 2025 | **Live:** gigishot-genai-blog.hf.space/

Developed an AI-powered web application that automatically generates blog content from user-provided topics using Large Language Models (LLMs). Integrated hosted LLM APIs with Flask to produce contextually relevant and structured articles. Implemented prompt engineering techniques to improve content quality and consistency.

Key Achievements:

- Automated blog generation workflow using LLM APIs.
- Designed effective prompts for high-quality content creation.
- Built a responsive web interface using Flask.
- Reduced manual content creation effort through AI-driven automation.

Technologies Used: Python, Flask, LLM APIs, Prompt Engineering, API Integration

Wordinator – Autocorrect Web Application | Python, Flask, Gradio, NLP

Nov 2024 – Dec 2024 | **Live:** gigishot-wordinator.hf.space/

Built an NLP-based autocorrect and text enhancement web application capable of detecting spelling errors and providing context-aware corrections. Developed an interactive Gradio interface for real-time user interaction.

Key Achievements:

- Implemented intelligent spelling correction using NLP techniques.
- Improved text quality through context-aware suggestions.
- Developed an easy-to-use web interface for instant feedback.
- Applied natural language processing concepts to real-world text data.

Technologies Used: Python, Flask, Gradio, NLP, Text Processing

Cat vs Dog vs Neither Image Classifier | Python, TensorFlow, Keras, CNN

Apr 2025 – Jun 2025 | **Live:** gigishot-cat-dog-neither-classification.hf.space/

Designed and trained an image classification model to categorize images into three classes: Cat, Dog, and Neither. Utilized transfer learning with a pre-trained Convolutional Neural Network (CNN) to improve performance and training efficiency.

Key Achievements:

- Trained and evaluated the model on 3,000+ images.
- Achieved **96% classification accuracy**.
- Applied transfer learning to reduce training time and improve accuracy.
- Performed image preprocessing, model tuning, and evaluation.

Technologies Used: Python, TensorFlow, Keras, CNN, Transfer Learning

Handwritten Word Recognition System | Python, TensorFlow, OpenCV, OCR

Dec 2025 – Present | **Live:** github.com/Gargie-ui/Handwritten-Word-Recognition-System-

Developing a handwritten word recognition system that converts handwritten text images into machine-readable text using Optical Character Recognition (OCR) and deep learning techniques. Leveraged CNN-based architectures and image preprocessing methods to enhance recognition performance.

Key Achievements:

- Built an end-to-end OCR pipeline for handwritten text recognition.
- Improved recognition quality by approximately **30%** through preprocessing and model optimization.
- Applied computer vision and deep learning techniques for text extraction.
- Enhanced model robustness across different handwriting styles.

Technologies Used: Python, TensorFlow, OpenCV, CNN, OCR, Deep Learning